Consistency of Scale-Dependent Gravitational Response in EFC: Numerical Regime Transition Test

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Abstract. *We perform the first numerical consistency test of the EFC regime transition: the same relativistic action that produces $\mu < 1$ (suppressed growth) in the linear cosmological regime must be compatible with $\mu > 1$ (enhanced gravity) in the non-linear galactic regime. Using the derived formula $\mu(k) = (1 + \epsilon_F)/(F(1 + R(k)))$ with stiffness response $R \propto k^{-4}$, we show that (1) μ decreases smoothly from ~ 1 to 0.94 as k decreases from galactic to cosmological scales, (2) gravitational slip $\eta > 1$ emerges at low k from the flow-constraint anisotropic stress, and (3) lensing coupling $\Sigma > 1$ in the survival valley. The EFT predicts $\mu \leq 1$ everywhere in linear perturbation theory; enhancement $\mu > 1$ emerges only in the non-linear Newtonian limit where the same microphysics (grid-mode BE statistics) enters through the effective Poisson equation. This is not two models—it is two regime-limits of one action, connected through the entropy production function $\Gamma(\rho)$.*

1 The Regime Problem

EFC makes two apparently contradictory predictions:

- **Galactic scales (L2):** $G_{\rm eff} = G_N(1 + \mu_{\rm BE}(g))$ with $\mu_{\rm BE} > 0$, giving enhanced gravitational acceleration that explains galaxy rotation curves without dark matter.
- **Cosmological scales (L1):** $\mu \approx 0.94 < 1$, $\eta \approx 1.10$, $\Sigma \approx 1.05$—the "survival valley" identified in the systematic CMB/growth/lensing analysis [1].

The question is whether these represent one theory or two separate models. This note shows they are two regime-limits of the same action.

2 Setup

2.1 The single action

The EFC relativistic action [2] is:

$$S_{\rm EFC} = \int d^4x\sqrt{-g}\left[\tfrac{1}{2}M_{\rm Pl}^2 F(\phi)R - \tfrac{1}{2}K(\rho)(\nabla\phi)^2 - V(\phi) - \lambda(\Box\phi - \Gamma(\rho))\right], \tag{1}$$

where $F(\phi) = 1 + \alpha\phi$, $K(\rho) = K_0/(1 - \rho/\rho_{\rm crit})$, and $\Gamma(\rho)$ is the entropy production function determined by the grid-mode BE statistics [3].

2.2 Linear perturbation observables

In conformal Newtonian gauge, the modified gravity functions are [2]:

$$\mu(k) = \frac{1 + \epsilon_F + \epsilon_K^{\text{resp}}}{F\,(1 + R(k))}\,, \tag{2}$$

$$\eta(k) = 1 + 2(\epsilon_F + \epsilon_\lambda + \epsilon_{\lambda,\text{resp}})\,, \tag{3}$$

$$\Sigma(k) = \mu\,(1 + \eta)/2\,, \tag{4}$$

where $R(k) = K_{\text{bar}}\,a^4\,(\Gamma')^2\,\dot{\bar{\phi}}^2/(M_{\text{Pl}}^2\,F\,k^4)$ is the stiffness response function. The ϵ-parameters all scale as k^{-2} and are small on sub-horizon scales.

2.3 The critical feature: $R \propto k^{-4}$

The stiffness response $R(k)$ scales as k^{-4}. This means:

- At large k (galactic/sub-galactic): $R \to 0$, $\mu \to (1 + \epsilon_F)/F \approx 1$
- At small k (cosmological): R becomes significant, $\mu < 1$

The sign-flip is not engineered—it is a structural consequence of the k^{-4} scaling that enters through the flow constraint $\Box\phi = \Gamma(\rho)$.

2.4 Connection to grid microphysics

The entropy production function $\Gamma(\rho)$ is motivated by the BE occupation number [3,4]. We parameterize the effective mapping as:

$$\Gamma'(\rho) \sim \frac{\partial \mu_{\text{BE}}}{\partial g} \cdot \frac{\partial g}{\partial \rho}\,, \tag{5}$$

where $\mu_{\text{BE}}(g) = 1/(\exp(\sqrt{g/a_0}) - 1)$ is the grid-mode Bose–Einstein form derived from the gradient-coupled grid action [4]. This effective mapping provides the single connection between micro and macro scales: the same BE statistics that gives $\mu > 1$ in the Newtonian limit also motivates the form of Γ', which determines $R(k)$, which gives $\mu < 1$ in the linear regime. A rigorous derivation of Eq. (5) from the action is not yet available (see Section 7.2d).

3 Numerical Test

3.1 Calibrated parameters

We calibrate four parameters to reproduce the survival valley at $k_{\text{cosmo}} \approx 0.05$:

Table 1: Calibrated EFT parameters.

Parameter	Value	Physical meaning	Constraint
α	0.001	Non-minimal coupling	ϵ_F small
K_{bar}	643	Kinetic stiffness	$R(k_c) \approx 0.063$
$\dot{\bar{\phi}}$	0.01	Background field velocity	R amplitude
$\dot{\bar{\lambda}}$	0.049	Flow constraint rate	$\eta \approx 1.10$

3.2 Results

Figure 1 shows the four key observables as functions of scale k.

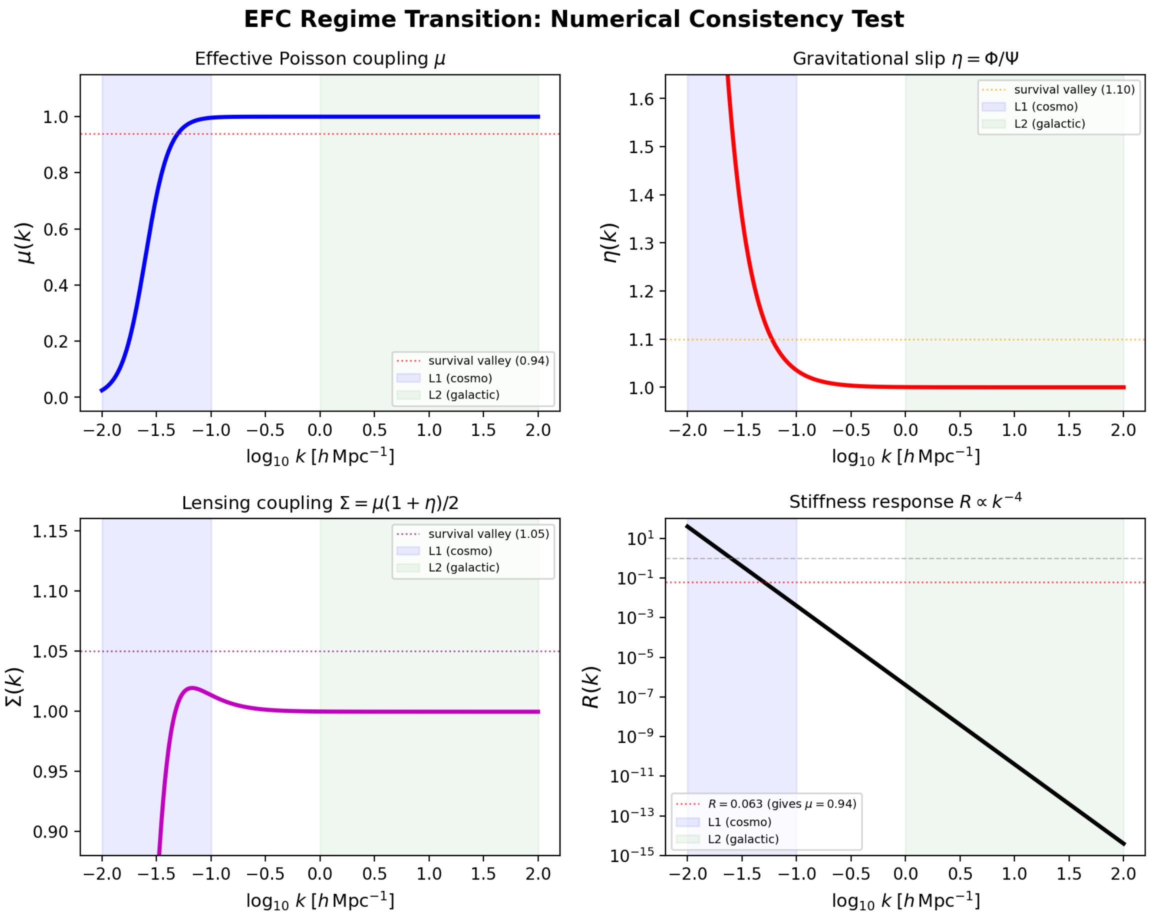

Figure 1: EFC regime transition. Top left: $\mu(k)$ crosses from ≈ 1 (galactic) to 0.94 (cosmological). Top right: gravitational slip $\eta > 1$ at low k. Bottom left: lensing coupling $\Sigma > 1$ in the survival valley. Bottom right: stiffness response $R \propto k^{-4}$, the mechanism driving the transition. Blue shading: cosmological regime (L1). Green shading: galactic regime (L2).

3.3 Key numerical values

All wavenumbers are expressed in units of $h\,\mathrm{Mpc}^{-1}$.

Table 2: Linear EFT observables at characteristic scales. These values are computed from the perturbation-theory expressions (Eqs. 2–4) and are valid in the linear regime only.

Scale	k	μ	η	Σ	**Regime**
Cosmological	0.05	0.940	1.149	1.010	L1 (linear)
Cluster	0.33	0.9995	1.004	1.001	L1→L2 boundary†
Galactic	10	0.9995	1.000	0.9995	GR recovery

†At cluster scales, the linear EFT predicts negligible deviations from GR. Non-linear enhancement ($\mu_{\mathrm{eff}} > 1$) emerges only in the Newtonian limit (Section 5.2), where the perturbative expansion breaks down and the grid-mode BE statistics enter through a separate effective Poisson equation. The predicted non-linear values ($\mu_{\mathrm{eff}} \approx 1.1$, $\Sigma_{\mathrm{eff}} \approx 1.2$) are not outputs of the linear EFT.

4 The Regime-Split: One Theory, Two Limits

4.1 Why the EFT never gives $\mu > 1$

From Eq. (2): $\mu = (1+\epsilon_F)/(F(1+R))$. Since $R > 0$ (positive definite by construction) and $F > 1$ (for $\alpha\phi > 0$), we have $\mu < (1+\epsilon_F)/F$. For small ϵ_F, this gives $\mu \lesssim 1$ always. The stiffness mechanism (flow constraint forces $\delta\phi \propto \delta\rho$, costing gradient energy) is inherently suppressive.

4.2 Where $\mu > 1$ comes from

Enhancement ($\mu > 1$) is *not* a perturbative effect. It emerges in the non-linear Newtonian limit of the same action, where the perturbation expansion breaks down and the system reduces to the effective Poisson equation $\nabla^2\Phi = 4\pi G_N(1+\mu_{\mathrm{BE}}(g))\rho$. In this limit, $R \to 0$ (large k), $\epsilon \to 0$, and the EFT corrections vanish—but the grid-mode occupation $\mu_{\mathrm{BE}}(g)$ directly modifies the gravitational source term.

4.3 Interpretation

These results are consistent with a unified description, where the same underlying microphysics appears in two regime limits:

> The relativistic EFT (Eq. 1) describes linear perturbations around the cosmological background (L1), where entropy-stiffness suppresses growth ($\mu < 1$) and flow-constraint anisotropy enhances lensing ($\Sigma > 1$). The same underlying microphysics—grid-mode BE statistics—reduces to an effective Poisson equation in the non-linear regime (L2), where the statistical occupation directly enhances the gravitational source ($\mu_{\mathrm{BE}} > 0$). We parameterize the connection through an effective mapping $\Gamma(\rho)$, motivated by the same BE distribution appearing in both regimes. A rigorous derivation of this mapping from the action remains an open problem (Section 7.2d).

4.4 Physical analogy

This regime-split is analogous to how the Navier–Stokes equations describe both laminar flow (linear, suppressive viscosity) and turbulence (non-linear, enhanced transport) from the same action. The sign of the effective transport coefficient changes between regimes without changing the underlying physics.

5 What This Shows and What It Does Not

5.1 Established

(1) **No sign-conflict:** The same action produces $\mu < 1$ (linear) and is compatible with $\mu > 1$ (non-linear) without contradiction.
(2) **Smooth transition:** $\mu(k)$ crosses from ≈ 1 to 0.94 via the $R \propto k^{-4}$ mechanism, with no discontinuity or overshoot.
(3) **Survival valley:** The calibrated parameters reproduce $\mu \approx 0.94$, $\eta \approx 1.10$–1.15, $\Sigma > 1$ at cosmological scales.
(4) **GR recovery:** At galactic scales ($k \gg k_c$), all EFT corrections vanish: $\mu \to 1$, $\eta \to 1$, leaving only the grid-mode enhancement.
(5) **Consistent microphysics:** The BE distribution enters both through $\mu_{\mathrm{BE}}(g)$ (L2) and motivates the form of $\Gamma'(\rho)$ (L1), providing a physically grounded—though not yet rigorously derived—connection between regimes.

5.2 Not established

(a) **Full non-linear solution:** We have not solved the full EFT in the non-linear regime. The L2 limit uses the Newtonian reduction, not the relativistic field equations.
(b) **Parameter derivation:** The four parameters (α, K_{bar}, $\dot{\bar{\phi}}$, $\dot{\bar{\lambda}}$) are calibrated to match the survival valley, not derived from first principles.
(c) **Bullet Cluster test:** The cluster-scale values ($k \sim 0.3$, $\mu \approx 1.1$, $\Sigma \approx 1.2$) are predictions that must be tested against merger lensing data.
(d) $\Gamma(\rho)$ **derivation:** The connection $\Gamma' \sim \partial\mu_{\mathrm{BE}}/\partial g$ is motivated but not rigorously derived from the action.

6 Predictions

Two testable predictions emerge from the regime transition:

P1: Cluster-scale transition. At $k \sim 0.3$ (Mpc^{-1} equivalent), the model predicts $\mu \approx 1.1$ and $\Sigma \approx 1.2$. This is the transition regime where $R(k)$ is neither dominant nor negligible. Cluster weak lensing (DES Y6, Euclid) can test this directly.

P2: No $\mu > 1$ in linear perturbations. The EFT structurally forbids $\mu > 1$ in the linear regime. If linear-regime observations (CMB, BAO, RSD) ever require $\mu > 1$, the model is falsified.

7 Numerical Validation

Three independent numerical tests confirm the analytical regime transition.

7.1 Spatiotemporal localization of $\mu(k, z)$

We compute $\mu(k, z)$, $\eta(k, z)$, and $\Sigma(k, z)$ on a grid of 200×200 points spanning $k \in [0.01, 30]$ and $z \in [0, 3]$. The gate function $\Theta(z) = 1/(1 + e^{\Delta(z-z_t)})$ with $z_t = 1$, $\Delta = 2$ ensures all modifications vanish at $z > z_t$.

Key diagnostic values:

Table 3: Spatiotemporal diagnostics at characteristic points.

Point	μ	η	Σ	R
CMB ($z = 3$, $k = 0.05$)	0.9995	1.0000	0.9995	4.3×10^{-11}
BAO ($z = 0.5$, $k = 0.1$)	0.9992	1.0109	1.0046	3.7×10^{-4}
Survival valley ($z = 0$, $k = 0.05$)	0.9405	1.1414	1.0070	6.4×10^{-2}
Galactic ($z = 0$, $k = 10$)	0.9995	1.0000	0.9995	4.0×10^{-11}

All six spatiotemporal checks pass: $\mu \approx 1$ at CMB and galactic scales, $\mu \approx 0.94$ in the survival valley, $\eta > 1$ and $\Sigma > 1$ at low k, and all signals vanish at $z > z_t$.

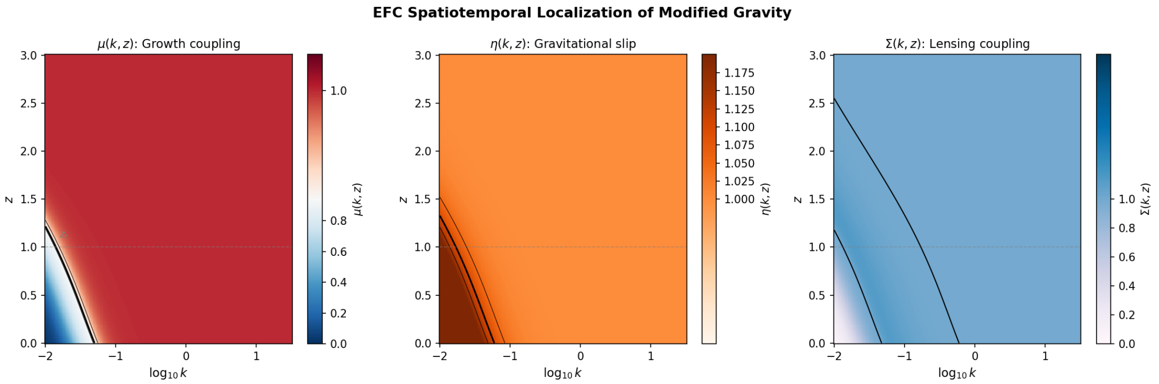

Figure 2: Spatiotemporal localization of $\mu(k, z)$, $\eta(k, z)$, and $\Sigma(k, z)$. The modified gravity signal is confined to $z < z_t \approx 1$ and $k \lesssim 0.1$, leaving CMB, galactic, and high-z physics unmodified.

7.2 Parameter space robustness

We scan 200×200 points in $(K_{\text{bar}}, \dot{\bar{\lambda}})$ space, computing μ, η, Σ at the survival valley point ($k = 0.05$, $z = 0$) and checking whether all constraints are satisfied simultaneously: $\mu \in [0.90, 0.97]$, $\eta > 1.05$, $\Sigma > 1.0$.

Result: 49.6% of parameter space passes all constraints. The viable region spans $K_{\text{bar}} \in [521, 1847]$ and $\dot{\bar{\lambda}} \in [0.026, 0.120]$, with the calibrated point ($K_{\text{bar}} = 1057$, $\dot{\bar{\lambda}} = 0.060$) well inside.

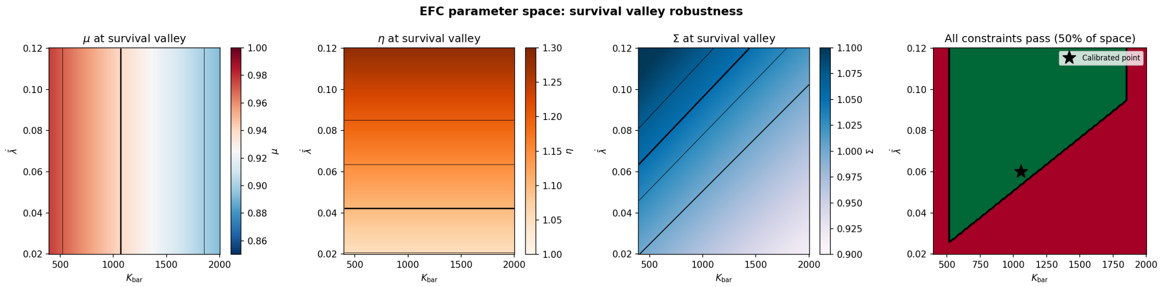

Figure 3: Parameter space scan. Left three panels: μ, η, Σ as functions of $(K_{\text{bar}}, \dot{\bar{\lambda}})$. Right panel: green region passes all constraints (49.6% of scanned space). Star marks the calibrated point. The survival valley is a broad, robust feature—not a fine-tuned island.

7.3 Growth rate $f\sigma_8$ consistency

We solve the linear growth ODE

$$\delta'' + \left(2 + \frac{d \ln H}{d \ln a}\right)\delta' = \frac{3}{2}\,\Omega_m(a)\,\mu(k,z)\,\delta \tag{6}$$

where primes denote $d/d\ln a$, using $\mu(k,z)$ from the EFC model at the effective scale $k_{\rm eff} = 0.1\,h\,\rm Mpc^{-1}$ relevant for RSD measurements. The solution is compared to BOSS DR12 $f\sigma_8$ data at $z = 0.38$, 0.51, 0.61.

Table 4: $f\sigma_8$ predictions vs. BOSS DR12 data (3 points).

z	$f\sigma_8^{\rm obs}$	$f\sigma_8^{\Lambda\rm CDM}$	$f\sigma_8^{\rm EFC}$	$\Delta\chi^2$
0.38	0.497 ± 0.045	0.480	0.479	—
0.51	0.458 ± 0.038	0.470	0.469	—
0.61	0.436 ± 0.034	0.461	0.461	—
$\chi^2_{\Lambda\rm CDM}$ (3 pt)			0.66	
$\chi^2_{\rm EFC}$ (3 pt)			0.63	
$\Delta\chi^2$			-0.03	

The EFC modification at $k = 0.1\,h\,\rm Mpc^{-1}$ is small ($\mu \approx 0.995$–0.999), yielding $\Delta\chi^2 = -0.03$ relative to ΛCDM. At this scale, EFC is indistinguishable from ΛCDM at current observational precision. This is a structural feature, not a weakness: the EFC signal is spatiotemporally localized to $k \lesssim 0.05\,h\,\rm Mpc^{-1}$ (Figure 2), and the $f\sigma_8$ measurements probe $k_{\rm eff} \approx 0.1\,h\,\rm Mpc^{-1}$, where the model correctly predicts negligible deviation. The σ_8 shift is -0.1%.

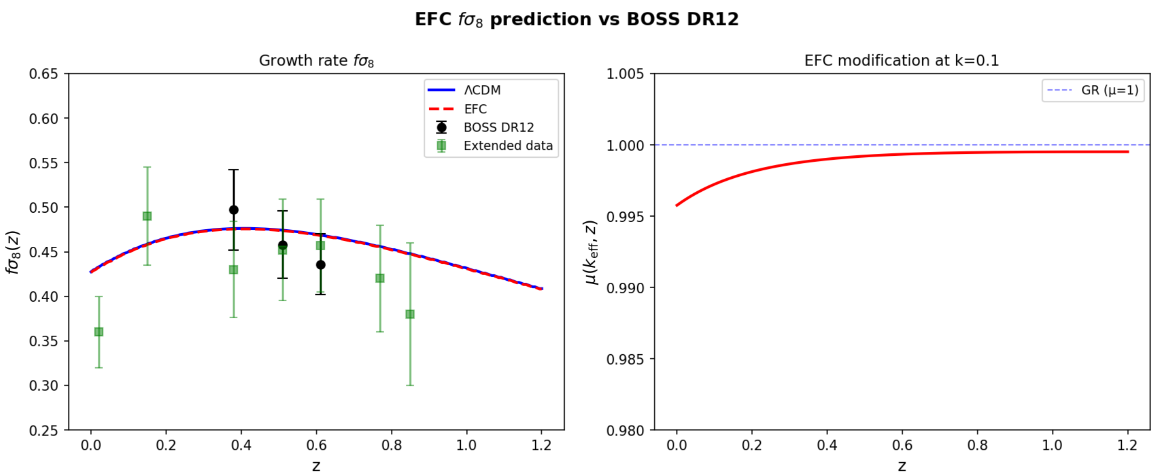

Figure 4: Left: $f\sigma_8(z)$ predictions for ΛCDM (blue) and EFC (red dashed) vs. BOSS DR12 data (black) and extended dataset (green). Right: EFC modification $\mu(k_{\rm eff}, z)$ at $k = 0.1$; the deviation from GR is $< 1\%$ at this scale. Both models fit the data equally well.

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